Tech Preparedness Guide

Digital tools, apps, and devices to keep you safe and connected in the backcountry

Pierce County 4x4 Search & Rescue | piercecounty4x4sar.org

Your smartphone is one of the most powerful safety tools you can carry into the backcountry -- but only if it's prepared. As a search and rescue unit, we've responded to calls where a fully charged phone with offline maps could have prevented the emergency entirely. This guide covers the apps, settings, and devices that can keep you safe when cell service disappears.

CRITICAL REMINDER

Technology is a supplement to -- not a replacement for -- the Ten Essentials. Always carry a physical map and compass. Batteries die, screens crack, and satellites lose signal. But a well-prepared phone dramatically improves your safety margin.

Recommended Navigation Apps

AllTrails

The most popular hiking app with 400,000+ trail guides, detailed maps, user reviews, photos, and elevation profiles. Great for discovering and planning hikes. Best for casual to regular hikers.

Free: Trail discovery, basic maps. AllTrails+ ($35.99/yr): Offline maps, wrong-turn alerts.

iOS: App Store

https://www.alltrails.com

Android: Google Play

Gaia GPS

The gold standard for serious backcountry navigation. USGS topo maps, satellite imagery, slope-angle shading, public/private land boundaries. This is what many SAR teams use.

Free: Basic maps + GPS tracking. Premium ($39.99/yr): Full offline topos, weather, slope analysis.

iOS: App Store

https://www.gaiagps.com

Android: Google Play

SAR PRO TIP:

Gaia GPS is widely used by search and rescue teams nationwide because of its detailed topo layers and ability to share GPS coordinates precisely. If you call 911 with Gaia coordinates, we can find you faster.

onX Backcountry

Excellent for understanding land ownership (public vs. private), hunting units, and backcountry access. Best for hunters, anglers, and mixed public/private land.

Premium ($29.99/yr): Full offline maps, 3D terrain, weather, land ownership layers.

iOS: App Store

https://www.onxmaps.com/backcountry

Android: Google Play

CalTopo

The best tool for trip planning. Create and print custom topo maps, plan routes with elevation profiles. Exceptionally powerful web interface for pre-trip research.

Free: Full web planning. Pro ($50/yr): Mobile app with offline maps + slope shading.

iOS: App Store
https://caltopo.com

Android: Google Play

Free Offline Maps (Already on Your Phone)

Google Maps -- Download Offline Areas

Google Maps lets you download large map areas for offline use. While it doesn't show trails in detail, it's excellent for road navigation to trailheads and as a backup.

How to download (iOS & Android):
1. Open Google Maps while connected to WiFi
2. Search for your destination area
3. Tap the location name at the bottom
4. Tap the three-dot menu > 'Download offline map'
5. Adjust the blue rectangle to cover your area
6. Tap 'Download'

Note: Offline maps expire after 30 days. Download fresh maps before each trip.

Apple Maps -- Download Offline Maps (iOS 17+)

Apple Maps now supports offline maps with turn-by-turn navigation, place info, and ETA -- all without cell service.

How to download:
1. Open Apple Maps
2. Tap your profile icon (bottom right)
3. Tap 'Offline Maps' > 'Download New Map'
4. Search for the area or zoom/pan to select it
5. Adjust the region and tap 'Download'

Tip: Apple Maps integrates with iPhone's crash detection and Emergency SOS via satellite.

Organic Maps

Completely free, open-source, privacy-focused offline maps built on OpenStreetMap. Hiking trails, elevation contours, cycling routes. No account, no ads, no tracking. Open the app, zoom to your area, and it auto-downloads.

100% free. No subscription. No account required.

iOS: App Store
https://organicmaps.app

Android: Google Play

SAR PRO TIP:
Download maps for your ENTIRE route, not just the trailhead. Include surrounding areas in case you need to navigate to a different exit point. Download at home on WiFi -- don't rely on trailhead cell service.

Phone Preparation Checklist

Before You Leave Home

- ❑ **Charge to 100%:** Start every trip with a full battery. Charge overnight before your trip.
- ❑ **Update your OS:** Make sure iOS or Android is up to date -- especially before relying on satellite SOS features.
- ❑ **Download offline maps:** Download maps for your entire route area in ALL the map apps you use.
- ❑ **Update Medical ID:** On iPhone: Health app > Medical ID. On Android: Settings > Safety & Emergency. Include allergies, medications, blood type, and emergency contacts.
- ❑ **Set up emergency contacts:** Add ICE (In Case of Emergency) contacts to your phone. These are shared automatically during SOS calls.
- ❑ **Enable location services:** Turn on GPS/Location Services. In an emergency, this lets 911 dispatchers and SAR teams locate you.
- ❑ **Test your apps:** Open each navigation app and verify your offline maps load correctly without cell service (turn on airplane mode to test).

Battery Conservation on the Trail

- ❑ **Low Power Mode:** Enable immediately when you start hiking. iPhone: Settings > Battery. Android: Settings > Battery > Battery Saver.
- ❑ **Reduce screen brightness:** Set to 30-40%. The screen is the biggest battery drain.
- ❑ **Close background apps:** Swipe away apps you aren't using. Social media apps drain battery even in the background.
- ❑ **Limit camera use:** Each photo session wakes the screen and processor. Take photos in batches, then lock your phone.
- ❑ **Airplane mode (use carefully):** Saves significant battery but disables cell and satellite connections. Only use if you have a separate satellite communicator. Otherwise, leave cellular ON so 911 can reach you.

IMPORTANT: AIRPLANE MODE WARNING

Turning on Airplane Mode disables your phone's ability to connect to cell towers AND satellites. If you have no other emergency communication device, DO NOT use Airplane Mode. A dead phone found at your location is more useful to rescuers than a charged phone that was invisible to the network. Only use Airplane Mode as a last resort to preserve battery for a critical 911 call.

Power Banks & Charging

A portable power bank is essential for any trip longer than a few hours. Cold weather, GPS use, and photography drain batteries fast.

Sizing Guide

Day hike (2-8 hours): 5,000 mAh -- one full phone recharge, weighs 4-5 oz.
Overnight (1-2 days): 10,000 mAh -- 2-3 full recharges, weighs 6-8 oz.
Multi-day (3-5 days): 20,000 mAh -- 4-5 full recharges, weighs 12-14 oz.

Look for: USB-C Power Delivery (PD) for fast charging, and an IP rating (IP67+) for water/dust resistance if you'll be in wet conditions.

Cold Weather Tips

Lithium batteries lose 20-40% capacity in cold temps. In winter:

- Keep your phone and power bank in an inside pocket, close to your body
- At night, put both in your sleeping bag near your core
- Charge devices in the evening while you still have body warmth
- A phone showing 0% in extreme cold may recover capacity when warmed up

Emergency Communication Devices

iPhone Emergency SOS via Satellite (iPhone 14+)

If you have an iPhone 14 or newer, you have satellite SOS built in -- no subscription required. When you try to call 911 with no cell service, your iPhone will offer to connect via satellite.

How it works:
1. Try calling 911 normally -- if no service, iPhone prompts satellite SOS
2. Point your phone toward the sky (follow on-screen direction indicator)
3. Answer the emergency questionnaire (injury type, number of people, etc.)
4. Your location, Medical ID, and battery level are sent to emergency services
5. Stay still -- rescuers are coming to the coordinates your phone transmitted

Requirements: Clear view of the sky. May not work under dense tree canopy or in deep canyons. Works in the US, Canada, UK, Europe, Australia, and Japan.

SAR PRO TIP:
Practice the satellite SOS demo BEFORE you need it: iPhone Settings > Emergency SOS > Try Demo. This lets you experience the connection process without sending an actual alert. Also update your Medical ID -- SAR teams receive this information automatically.

Android Satellite SOS (Pixel 9+, Samsung Galaxy S25+)

Select Android phones now support satellite emergency messaging. On supported devices, when you have no cell signal and try to contact 911, the phone will offer satellite connectivity. Setup and usage varies by device -- check your phone's settings under Safety & Emergency.

T-Mobile Satellite via Starlink

T-Mobile offers satellite connectivity via Starlink on 60+ phone models (iPhone 13+, Galaxy S21+, Pixel 9+). Satellite 911 texting is FREE for all carriers -- AT&T, Verizon, anyone -- for emergency use only. Full satellite messaging (non-emergency texts, apps, data) is $10/month add-on, or included with Go5G Next and Experience Beyond plans. Sign up BEFORE your trip:

Sign Up for Free Satellite 911 Texting
t-mobile.com/coverage/satellite-phone-service/911-texting-signup

Dedicated Satellite Communicators

For regular backcountry users, a dedicated satellite communicator is the gold standard. Unlike phone-based satellite SOS, these allow two-way messaging and work under tree canopy.

Garmin inReach Mini 2 (~$400, plans from $14.95/mo) -- Two-way satellite texting, SOS with 24/7 GEOS monitoring, GPS tracking family can follow, 14-day battery.

Garmin inReach Mini 2
garmin.com/en-US/p/765374

Zoleo Satellite Communicator (~$200, plans from $20/mo) -- Two-way messaging via satellite/cell/WiFi, SOS with GEOS monitoring, check-in button.

Zoleo Satellite Communicator
zoleo.com

ACR Bivy Stick (~$350, plans from $19.99/mo) -- Two-way Iridium messaging, SOS alerting, weather forecasts, location sharing.

ACR Bivy Stick
acrartex.com/products/bivy-stick

SAR PRO TIP:
If you hike regularly in areas without cell service -- including much of the Cascade Mountains and Mt. Rainier foothills -- a Garmin inReach pays for itself the first time you need it. Two-way messaging means we can coordinate with you during a rescue, ask about injuries, and give you instructions. That's not possible with a PLB or phone SOS.

Quick Reference: Pre-Trip Tech Checklist

- ❑ Phone charged to 100%
- ❑ Portable power bank charged to 100%
- ❑ Offline maps downloaded for entire route area
- ❑ Navigation app tested in airplane mode
- ❑ Medical ID updated (allergies, meds, emergency contacts)
- ❑ Emergency contacts set up on phone
- ❑ Location services enabled (GPS on)
- ❑ Latest OS update installed
- ❑ Satellite SOS demo practiced (iPhone 14+ / Pixel 9+)
- ❑ Weather forecast checked in app
- ❑ Trip plan shared with emergency contact
- ❑ Physical map and compass packed as backup

Check Cell Coverage Before Your Trip

Most Cascade mountain trails and Mt. Rainier backcountry have NO cell service. Check your carrier's coverage map before your trip:

T-Mobile Coverage Map
t-mobile.com/coverage/coverage-map

AT&T Coverage Map
att.com/maps/wireless-coverage.html

Verizon Coverage Map
verizon.com/coverage-map

FROM OUR TEAM TO YOU

We've responded to emergencies where a prepared phone saved a life, and emergencies where a dead phone made everything harder. The 15 minutes you spend preparing your tech before a trip is the easiest safety investment you'll ever make. Download the maps. Charge the battery. Tell someone your plans. Come home safe.